Data Conversion between CAD and GIS in Land Planning

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***Abstract*—In land planning, data conversion between CAD and GIS becomes the data-sharing a great challenge. In this paper, three aspects were studied. First of all, the organization style between CAD and GIS data are compared. Secondly, the data conversion method based on FME is proposed, and, we analyze its semantic transformation model system. Finally, the coordinate transformation which takes place before the data conversion is studied, as well as spatial elements deformation because of it. In conclusion, different organization style is an obstacle to data conversion between CAD and GIS. As for coordinate transformation, we program and expand its attribute with the help of LISP files in AutoCAD. The data conversion based on FME makes the expression of spatial information maximum inherited. The semantic transformation model of FME is designed to build the mapping relationship between point, line, area and corresponding text during the transformation process between source and destination data.**

Keywords- data conversion; CAD; GIS; land planning

I. INTRODUCTION

With the rapid development of computer technology and the increasing demand of spatial data for people, the data format which is used to describe a sort of spatial item is diversified development. In order to make use of available data effectively, and realize data sharing and interaction, it's particularly important to realize the conversion between different data formats.

AutoCAD, as traditional CAD software, which has powerful drawing and vector editing functions, is used in land and planning departments widely. From the perspective of geo-spatial information, however, AutoCAD is weak in expressing and analyzing the relationship of spatial data. In contrast, as a mainstream GIS software, ArcGIS can not only handle graphic data, but also model, analyze and manage the existing spatial data, meanwhile, it's able to gather geo-spatial factors to process and express in a unified model framework [1]. Because of its convenience in handling geo-spatial data, ArcGIS is accepted by more and more GIS enthusiasts.

Currently, most areas are carrying the new round of land use planning, such as Suzhou(China)、Nantong(China)、Wuxi (China)and so on. These places are required to build a database with ArcGIS software. However, the land use planning is always lagging behind the urban planning. So how to use the existing urban planning data serving for land use planning, which becomes an urgent problem to be solve present. Meantime, the data format conversion between CAD and GIS becomes a prominent problem for data sharing. Based on this background, we particularly describe the method of conversion between the two data format in land use planning, and give corresponding solution to the problems happened during data conversion.

II. DATA ORGANIZATION STYLE

A. CAD Data Organization Style

AutoCAD stores graphics data with DWG (graphic files) and DXF (graphic switching files) format. DWG is binary format file, which mainly consists of five parts: head, entity part, table part, block entity part and emergency entity head. The lengths of head and emergency entity head are fixed, and other three parts are different with the variation of entity [2].

The head of DWG file stores file marks, version information, index address and other important information; the entity part keeps all graphic entities, including point, line, arc, circle, size, mark etc.; the table part is designed for searching, which includes block table, layer table, font table, linear table, etc.; the block entity part is introduced for reducing the length of graphic files, each block brings a block table in order to search conveniently when it's created; the emergency entity head is used to prevent important head index information from damaging, and save a copy of the head at the same time.

A DWG file is also called an AutoCAD database [3], which is mainly used for storing graphic files. During the process of object-oriented programming, we can extended the attribute of AutoCAD through the program statements SetXData and GetXData, with the help of these statements , object's attribute can be connected. Generally speaking, the data of urban planning is DWG format, and it consists of multiple layers, while each

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layer has the same attributes. For example, road consists of simple lines and has length and width, then the same attributes item will gather together, so does the construction land. The separation management mode of layers provides us with great convenience when we deal with AutoCAD data and make a choice [4].

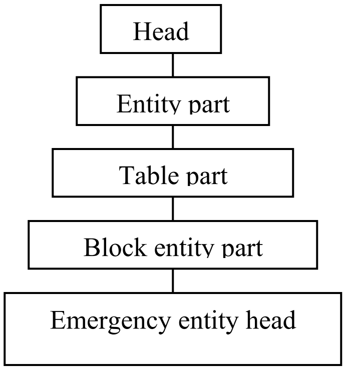

Figure 1. Structure of a DWG file

B. GIS Data Organization Style

Currently, mainstream GIS software is ArcGIS which is developed by ESRI Corporation. It contains powerful space analysis and 3D analysis function. The data formats of ArcGIS vary, shapefile, coverage, E00, Geodatabase (ArcSDE, MDB) and SDE are often used. On the process of land use planning mapping, the main data format is shapefile, therefore, this paper emphasizes on it.

Shapefile is an example of geographical data model, and it stores the spatial data and attributes in the same system. Shapefile stores two basic documents with geometry properties: one is the geometry characteristics of space factors, which is in the form of .SHP files; the other is the index of the spatial factors, which is in the form of .SHX files [5]. The former is also called key file, and the latter is called index file. In addition, Shapefile has a dBASE data table file, which exists with the suffix of .DBF, and it stores the attribute information of graphic elements.

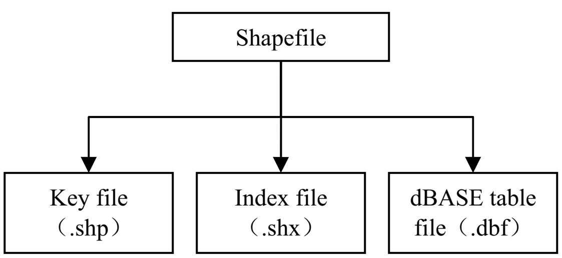

Figure 2. Structure of a shapefile

In a shapefile, the key file can be accessed directly, and its length is also varied, meanwhile each record in key file is described with a series of points. The index file keeps the offset of the corresponding objects from the head of the key file. The dBASE data table file contains the attribute of each record, which is corresponded with the key file.

C. Difference

There are four different aspects between CAD and GIS data, including data structure, data organization, topology and coordinate system.

From the view of data structure, CAD and GIS have their own data storage format, and the hierarchical system of each format is different. The main data structure of CAD is DWG file, while GIS has shapefile, coverage, Geodatabas and so on.

From the view of data organization, CAD and GIS adapt hierarchical management model. However, the former just joins multiple layers together simply, and each layer is independent, while the latter links every layer by its attribute.

From the view of topology, GIS data always contains topology among space entities, which is used to describe the relationship of spatial adjacent, inclusion and intersection, while CAD data cannot realize it.

From the view of coordinate system, CAD and GIS have great difference. The coordinate system of GIS is determined by datum plane and projection. Different countries have their own datum plane, in addition, the choice of projection is also various. CAD only supports two coordinate systems, one is World coordinate system (WCS), and the other is User coordinate system (UCS).

III. DATA CONVERSION

Land planning must be accord with urban planning in China, that means we can make use of the fruits of urban planning, such as the planning scopes of future construction land, or planning road lines. In urban planning, AutoCAD software is the main mapping tool. During the new round of land use planning in China, ArcGIS software is required. Therefore, DWG files in CAD will be converted to shapefile in GIS.

In urban planning, there are various forms of land elements. Commonly, the construction land and road line are what we need in land planning. The former is shown in the form of double or single line in AutoCAD, and the graphic is closed, meanwhile, there is no attribute. The latter has a third form, which is used to show the curve of a road with arc, and the road line is discontinuous, what's more, there is also no attribute description [6]. In this paper, data conversion of the plane objects will be introduced emphatically.

A. Traditional Data Conversion Mode

Traditional data conversion is finished through a narrow channel which is single, and this technique requires the same model between input data and output data [7]. As the development of data sharing technology, a new method of data conversion which is based on the public data format appears. In order to realize data sharing, CAD and GIS software defines their own public data formats. The public data format of AutoCAD is DXF file, and ArcGIS is E00 file. In order to achieve data conversion between CAD and GIS, one should convert CAD data into the public data format DXF first, and then to E00 which is the public data format of GIS, finally, we can convert E00 file into objective data type with GIS software. The model of data conversion above cannot realize data sharing before undergoing three operations, and the process of conversion is relatively complex, what's more, the data information easily loses after converting. The conversion above emphasizes on format, and ignores semantic change, so it cannot fully express the meaning of metadata after converting. Therefore, the conversion method is being replaced by newer and more convenient way.

B. Data Conversion Based on FME

In order to make spatial information maximum inherited and realize data sharing based on semantic level, we put forward a method of data conversion based on FME. FME is developed by Safe Software Corporation, which is used to convert spatial data. It is based on the principle of "semantic mapping", which provides the function of reconstruct data during the process of data conversion. Because of this, it can realize data conversion among 100 different spatial data formats [8].

FME will establish the mapping relationship between source and destination data during the process of data conversion [9], which is shown in the form of data flow. During the conversion process, the source data is CAD (DWG graphic file), and the destination is GIS (shapefile), their mapping table will be established by choosing corresponding data format. The model structure of FME is shown in Fig. 3.

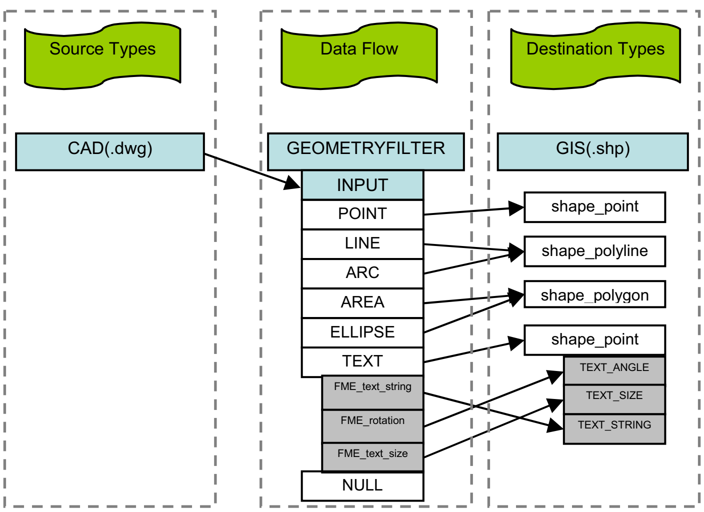

Figure 3. Model of FME

Normally, the source data is single, while the destination data is multiple, which means that they have a relationship of one-to-many. According to the model of FME, each DWG file can be divided into point, line, arc, area, ellipse and text, which turns into the corresponding destination types in the end. Point will be converted into shapefile point, line and arc will be changed to shapefile polyline, similarly, area and ellipse will be converted into shapefile polygon, text will be changed into shapefile point, whose geometric attributes remain unchanged.

In land planning, the area which represents construction land and the road line are needed. So the work of data conversion turns out to realize the change of area (or ellipse) and line (or arc). In addition, the conversion of text could remain the attribute of space entity.

IV. Coordinate Transformation

A. Space Coordinate Transformation

The software of AutoCAD and ArcGIS are independent, and they are different in locating space objects. The coordinate in AutoCAD is the result of spherical coordinate which projects on a datum level. Under the toolbar of view, AutoCAD owns world coordinate system (WCS) and user coordinate system (UCS). ArcGIS also has two sets of coordinate system, one is geographic coordinate system (GCS), and the other is a projected coordinate system (PCS).

In urban planning, the CAD data adapts world coordinate system. However, in the new round of land use planning in China, the coordinate system is required to use Gauss Kruger projection. The geographic coordinate system should be Beijing 54 or Xian 80 coordinate system. The first problem during data conversion is coordinate system transformation. In this paper, we take the transformation which is from world coordinate system to Beijing 54 coordinate system for example.

In AutoCAD software, we can expand the attribute of one layer by programming with the help of LISP files .In order to realize the transformation of two different coordinate systems, we should obtain several parameters, such as the rotation angle of two different coordinate systems, the transformation parameters and the offset from the original point. The formula of coordinate transformation is shown as follows:

$$\begin{aligned} x^{'} &= x_0 + k*(x*\cos\varphi - y*\sin\varphi) \\ y^{'} &= y_0 + k*(x*\sin\varphi + y*\cos\varphi) \end{aligned} \quad (1)$$

In the formula, $x^{'}$ and $y^{'}$ represent the coordinate in Beijing 54 coordinate system. x_0 and y_0 represent the offset between world coordinate system and Beijing 54 coordinate system. φ is the rotation angle between two coordinate systems. k is a transformation parameter, whose value is a constant.

According to the formula above, we program and make a new LISP file named "zb54" in AutoCAD. After creating the procedure, we should store the file under the subfolder called "support" which locates in the installation directory of AutoCAD. Finally, we could load the program in AutoCAD to realize the coordinate transformation. The command is as follows: (load "zb54"), zb54. Fig. 4 shows the change of coordinate range of a new-built land in land planning after coordinate transformation.

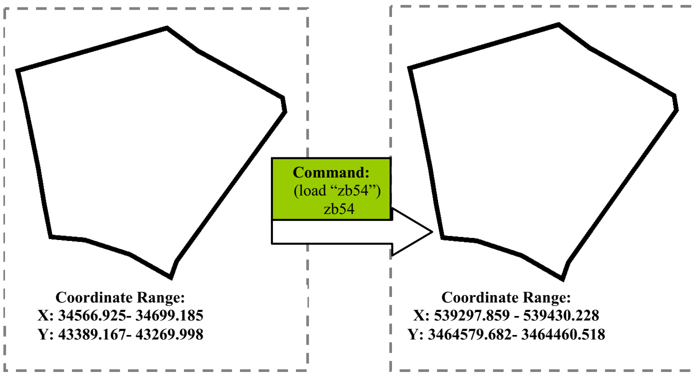

Figure 4. Comparion of a new-built after coordinate translation in land planning

B. *Spatial Element Deformation*

The polygon features in AutoCAD are not closed completely, so it will be out of shape after coordinate transformation. The deformation elements cannot be used in land planning because it is different from the original. If we still use the deformation data, there will be a great deviation when we calculating the area of the land in land planning. As for the deformation elements, the only way we can do is to edit secondly in AutoCAD and then make the conversion. Fig. 5 shows the deformation of a new-built land after coordinate transformation.

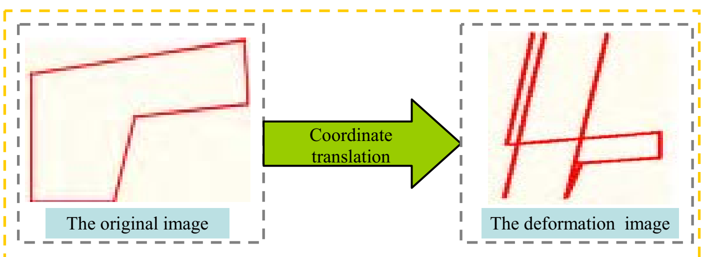

Figure 5. Deformation of a new-built after coordinate translation

How to solve the problem of deformation? How to avoid this situation during coordinate translation? The convenient solution lies in AutoCAD software. Before coordinate translation, we use the command called "BO", whose function is to convert the closed border of a graphic into multiple lines, and then rebuilds a surface. After the operating of "BO", the polygon becomes more stable, and cannot deform with other operating of the graphic. Therefore, if we delete the original graphic and keep the polygon generated by the operating of "BO", the problem of element deformation will not happen again, and we can go on carrying out the coordinate transformation. In addition, we can obtain polygon surrounded by lines with the operating of "BO". It's good for us to statistic the area of the road lines in land planning.

V. CONCLUSIONS

Land planning indicates the direction of development in a region, and plays an important role in regional strategy. However, it is often lagging behind urban planning. In order to use the fruits of urban planning, data conversion is an inevitable problem between CAD and GIS.

This paper emphasizes on the conversion style between CAD and GIS data, as well as coordinate transformation and element deformation in the process of data conversion. In conclusion, different organization style between CAD and GIS data is a challenge for data conversion. Data conversion means coordinate transformation at the same time, because CAD and GIS have their own coordinate systems. We achieve coordinate transformation by programming with the LISP files in AutoCAD. In this paper, we take an example of the transformation between WGS84 and Beijing54 coordinate system. In order to discriminate the accuracy of the transformed data, the manual interpretation and public participation are also used after data conversion. Data conversion based on FME makes the expression of spatial information maximum inherited, and achieves data-sharing of the semantic-level. The semantic transformation model of FME is designed actually to build the mapping relationship between point, line, area and the corresponding text during the data conversion process between source and destination. The effective method to avoid element deformation is to operate the command of "BO" before coordinate transformation.

This study makes convenience for the sharing of the two-data types and improves the cartographic efficiency in land planning.

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